

Class 3 SP250: Modular Arithmetic

The Last Digit

1. Find the last digit of $13215 + 99382$.

Answer: 7 because $5 + 2 = 7$.

2. Find the last digit of $13215 * 99382$.

Answer: 0 because $5 * 2 = 0$.

3. Find the last digit of 99382^{13215}

Answer: It is 8. Why?

Last digit of 99382^{13215} equals the last digit of 2^{13215} .

When in doubt, try to find the pattern:

2, 4, 8, 16, 32, 64, ...

So, the last digit repeats every four exponents. Since 13215 gives remainder 3 when divided by 4, we conclude that the last digit we seek is 8.

Definition: Modulo Let a, b , and m be integers with $m \neq 0$.

We say that $a \equiv b \pmod{m}$ (pronounced “a is congruent to b modulo m”) if $m \mid (a-b)$.

In other words, a and b are congruent modulo m if and only if a and b **have the same remainder** when divided by m .

Example: $17 \equiv 12 \equiv 7 \equiv 2 \pmod{5}$

Definition: Residue If $a \equiv r \pmod{m}$, r is a residue and can have values $0, 1, \dots, (m-1)$

Example: if $m = 5$, then possible residues are $0, 1, 2, 3, 4$

True or False?

1. $17 \equiv 5 \pmod{6}$.

True because $17 - 5 = 12$, which is divisible by 6. **$6 \mid 17 - 5$**

2. $5 \equiv 5 \pmod{12}$.

True because 0 is divisible by anything. **$12 \mid 5 - 5$**

3. $13 \equiv 7 \pmod{12}$.

False because 12 does not divide $(13 - 7) = 6$. **$12 \nmid 13 - 7$**

Mods - what's the big deal? So far the examples have been trivial and we haven't needed the definition of mods.

Question 1: What are the last 3 digits of $233992 * 232612$?

We want to find $233992 \times 232612 \pmod{1000}$

$$233992 \equiv -8 \pmod{1000}$$

$$232612 \equiv 612 \pmod{1000}$$

$$-8 \times 612 = -4896 \equiv 104 \pmod{1000}$$

Further explanation:

$$X \equiv 233992 * 232612 \equiv (-8) * 612 \equiv -(8 * 612) \equiv -(4896) \pmod{1000}.$$

$$X - (-4896) \text{ ends in } 000.$$

$$X + 4896 \text{ ends in } 000.$$

$$X = 104 \pmod{1000}.$$

Mod Manipulation Theorem

Let a, b, c, d and m be integers with $m \neq 0$.

- a) If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then $(a+c) \equiv (b+d) \pmod{m}$

Remainders ADD.

- b) If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then $(a-c) \equiv (b-d) \pmod{m}$

- c) If $a \equiv b \pmod{m}$ then $ac \equiv bc \pmod{m}$

- d) If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then $ac \equiv bd \pmod{m}$

Remainders MULTIPLY.

Examples:

1. $15 \equiv 3 \pmod{12}$ $19 \equiv 7 \pmod{12}$
 $15 + 19 \equiv 3 + 7 \pmod{12}$
 $34 \equiv 10 \pmod{12}$ ✓
2. $11 \equiv 2 \pmod{9}$ $7 \equiv 7 \pmod{9}$
 $11 \times 7 \equiv 2 \times 7 \pmod{9}$
 $77 \equiv 14 \pmod{9}$
check: $77 - 14 = 63$ which is divisible by 9 ✓

Mods, in general, cannot be canceled by division.

Example: $3x \equiv 3y \pmod{9}$. Does it imply that $x \equiv y \pmod{9}$? NO.

$x = 1$. $y = 4$. $3x = 3$. $3y = 12$. **$3 \equiv 12 \pmod{9}$?** ← YES.

Is $1 \equiv 4 \pmod{9}$? NO.

Proving the Mod Manipulation Theorem for Addition:

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then $(a+c) \equiv (b+d) \pmod{m}$

Proof: Use the definitions.

$a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ means that $m \mid (a-b)$ and $m \mid (c-d)$.

Then $m \mid (a-b) + (c-d) = (a+c) - (b+d) \Rightarrow (a+c) \equiv (b+d) \pmod{m}$ by definition.

Homework 2 Problem 4 revisited:

Find all primes p such that $p^2 + 2$ is also prime.

Solution with mods: Answer: $p=3$ only.

If p is not equal to 3, then $p \equiv 1 \text{ or } 2 \pmod{3}$.

Note that $p^2 + 2$ will be congruent to $1^2+2 = 3$ or $2^2 + 2 = 6$. Thus if p is not 3 then p^2+2 is a multiple of 3.

★ Perfect squares, when divided by 3, only give remainders 0 or 1!

This is the second highly non-trivial fact that you encounter in this class.

Every integer is either 0, 1, or 2 $\pmod{3}$.

$0 * 0 = 0$, $1 * 1 = 1$, $2 * 2 = 4 \equiv 1 \pmod{3}$.

Question 2: There are 9 1's written on the board. Jeffrey comes and inserts a single 2 anywhere in 10 possible spots (before the first 1, between 1st and 2nd, ...). Is it possible that Jeffrey's new 10-digit number is a square?

Answer: this is impossible.

A number, consisting of 9 1's and a 2, has a remainder 2 when divided by 3.

Because the sum of the digits of said number is 11, and 11 is congruent to 2 when divided by 3, hence not a square.

Question 3: What is the last digit of 7^{2022} ?

7, 49, 3, 1, 7, 9, 3, 1, 7, 9, 3, 1,

The pattern repeats with a cycle of 4.

$2022 \equiv 2 \pmod{4}$. So 7^{2022} ends with the same digit as 7^2 , 9.

Question 4: Nathan wrote a 2-digit number on the board. Tappan took Nathan's number and reversed the digits. What are all possible values of the absolute value of the difference between Tappan's number and Nathan's number?

Let's say Nathan's number is $\overline{ab} = 10a + b$.

Tappan's number is then $\overline{ba} = 10b + a$.

Their difference is $|(10a + b) - (10b + a)| = |9a - 9b| = 9|a - b|$.

The answer is all multiples of 9, starting from 0 and ending on 81.

Is this a complete proof?

No, it is not – in order for this to be a complete proof, one needs to demonstrate that all multiples of 9 are actually possible. By taking Nathan's number to be 90, 91, 92, ..., 99, we see that they indeed are all possible.

Divisibility by 9

What is the remainder when 10^{100} is divided by 9?

Answer: 1. $10^{100} = 10 * 10 * \dots * 10 \equiv 1 * 1 * 1 * \dots * 1 = 1 \pmod{9}$.

What is the remainder when 10^n is divided by 9 for some non-negative integer n ?

Answer: 1.

Divisibility rule by 9: Is 98376327 divisible by 9? How do you know?

It is because $9 + 8 + 3 + 7 + 6 + 3 + 2 + 7$ is a multiple of 9.

Question: Let n be a positive integer. Prove that the remainder when n divided by 9 can be computed as adding up all the digits of n and then finding the remainder of that sum modulo 9.

For example: $129834 \rightarrow 1 + 2 + 9 + 8 + 3 + 4 = 27 \equiv 0 \pmod{9}$.

Proof: Let $a_k, a_{k-1}, \dots, a_0$ be the digits of n .

$$n = \overline{a_k a_{k-1} \dots a_0}$$

$$n = a_k \cdot 10^k + a_{k-1} \cdot 10^{k-1} + \dots + a_1 \cdot 10 + a_0$$

Recall that $10^m \equiv 1 \pmod{9}$ for all m .

$$n \pmod{9} \equiv a_k \cdot 1^k + a_{k-1} \cdot 1^{k-1} + \dots + a_1 \cdot 1 + a_0$$

$$n \pmod{9} \equiv a_k + a_{k-1} + \dots + a_1 + a_0$$

This is adding up all the digits of n , just like we hoped to prove.

So the divisibility rule for 9 is, add up all the digits of the number, and if they sum to a multiple of 9, then the number is divisible by 9.